

things down to single measures that they can try to change. In America, the key measure used is typically the speed of a journey. The logic is that if people can get from A to B faster, then they have more time left over to work, or spend, or whatever. They can also go farther distances, meaning that more people can realistically reach any given job. When a lot of people already drive, this produces a logic that always favors widening roads, or building new ones, because it speeds up the time spent in traffic. And time spent in traffic is straightforwardly time wasted.

Following this logic, lots of firms try to estimate the costs of traffic congestion. In America, INRIX produces estimates of how congested various cities are; TomTom, a Dutch firm that sells satellite navigation technology, produces its own index of congestion, looking at 416 cities. The most congested cities on Earth are generally in poorer places—Mumbai, Istanbul, and Bangkok all feature highly on TomTom's list. But some richer cities come up fairly high too, such as Moscow, Paris, and London, as well as Los Angeles and New York. INRIX estimates, for example, that congestion costs the US \$88 billion per year, or around \$1,500 per driver. In the UK they put the figure at £21 billion, an even more staggering figure than the US one, given that the UK's economy is so much smaller. From this, we might conclude that there is not enough road space for people to get around, and that the most congested cities in particular need more of it. That's what Elon Musk has concluded, for sure.

But what the reaction to road tolling—whether it is that bridge in Kentucky or the M6 Toll—shows is that these estimates do not really stand up. The actual "cost" of congestion is what people would pay to avoid it. If so many people refuse to pay even a relatively small toll to avoid being trapped in a traffic jam, then that suggests in reality, their productivity is not really being damaged anywhere near as much as the estimates suggest. Perhaps their journeys were not actually that important to begin with, or else there were perfectly good alternatives they were choosing not to use. This is an example of what economists called "revealed preference." It applies a lot to transport. Poll people and they say that they want wider roads to reduce congestion. And yet if you actually build those roads, you discover that they will not pay for them—their preference is revealed, and either people don't actually mind congestion, or they would prefer not to drive.

And so trying to solve traffic congestion by endlessly building more roads is not a good use of resources. There are almost no big cities on Earth where traffic congestion is not at least an occasional problem. For sure, in a few cities, the population has declined so much and the road space available is so high that almost no road ever gets jammed. Detroit, for example, is wonderfully easy to drive through, at least outside of downtown, because almost nobody else is there. But congestion is, to a point, a symptom of success. If you have it, it means that people live in your city and want to get around it. And it is people, not cars, that need to move around cities.

So the question is, when faced with congestion, what should we do about it? In general, the best way to reduce congestion is to charge a reasonable price for using the roads. If the price of using a road is put at a level where the people who need them most, and have no other realistic way to get around, can afford to reasonably use them, and there is still crippling congestion, then roads are probably in short supply. But if, as with the bridge in Kentucky, the result of charging is that congestion falls significantly, then you already had enough road space to begin with, or possibly too much. And even if it does not, you can then invest the money raised in improving public transport options, so that they do not need cars anywhere near as much. And would we not be better off trying to shift some of the traffic off the roads onto other forms of transport?

And yet that is almost never how city engineers and politicians think. As Jeff Speck, an urban planner, puts it, induced demand is "the great intellectual black hole in city planning, the one professional certainty that everyone thoughtful seems to acknowledge, yet almost no one is willing to act upon."

The reason is because it is so bloody hard to think otherwise. Traffic congestion is one of those truly democratic things, in that it ruins people's lives all over the world. Whether you live in Manhattan or Mumbai, Jakarta, or Jacksonville, it sucks sitting in traffic. Even Elon Musk, who owns two private jets, knows it. And when you are sitting in a car, surrounded by other cars, the idea that a wider road would make things easier makes intuitive sense. It happens to be nonsense though. What would actually help you move along faster is if the other drivers were not there. And a relatively small toll is far more likely to achieve that than a wider road. When we give people extra road space for free, we are encouraging them to get around in a way that they would

* Reason Murphy vetoed toll hikes?

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